SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 02.1

Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter Using MATLAB.

Name: Dhinakaran. S **Reg. No.:** 192412353

Aim

To design a Chebyshev Type-I IIR low-pass filter and evaluate its frequency, impulse, group-delay, and pole-zero characteristics.

Procedure / Calculations

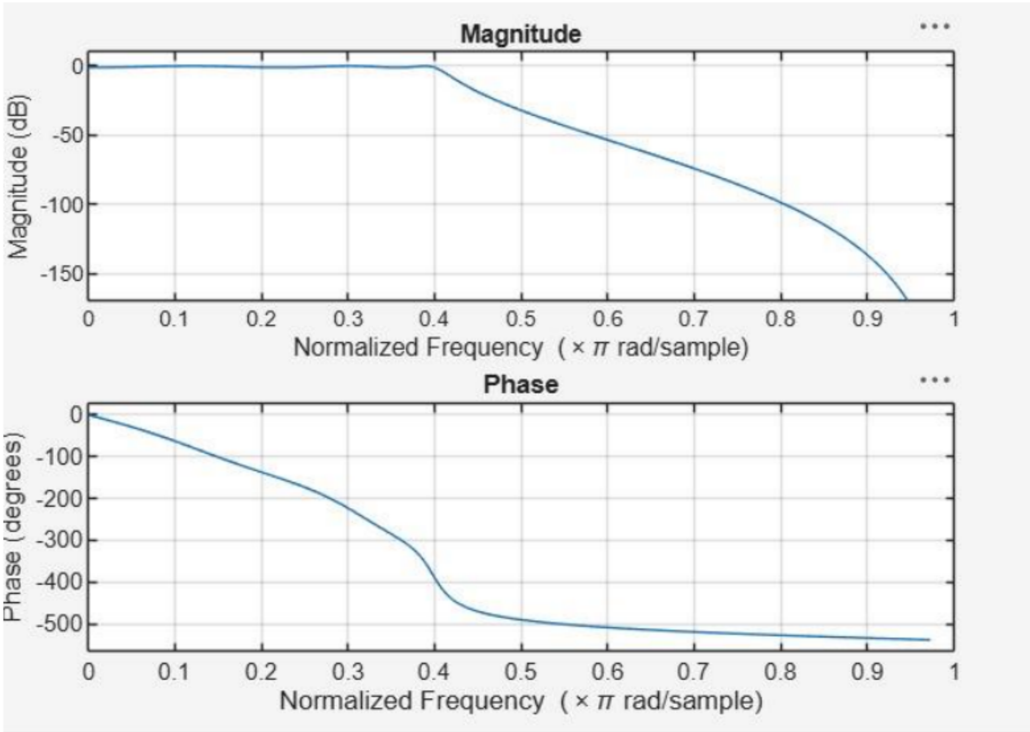
Set normalized passband/stopband edges, determine order with cheb1ord(), design with cheby1(), display order and cutoff, and verify stability from the pole locations.

MATLAB Program

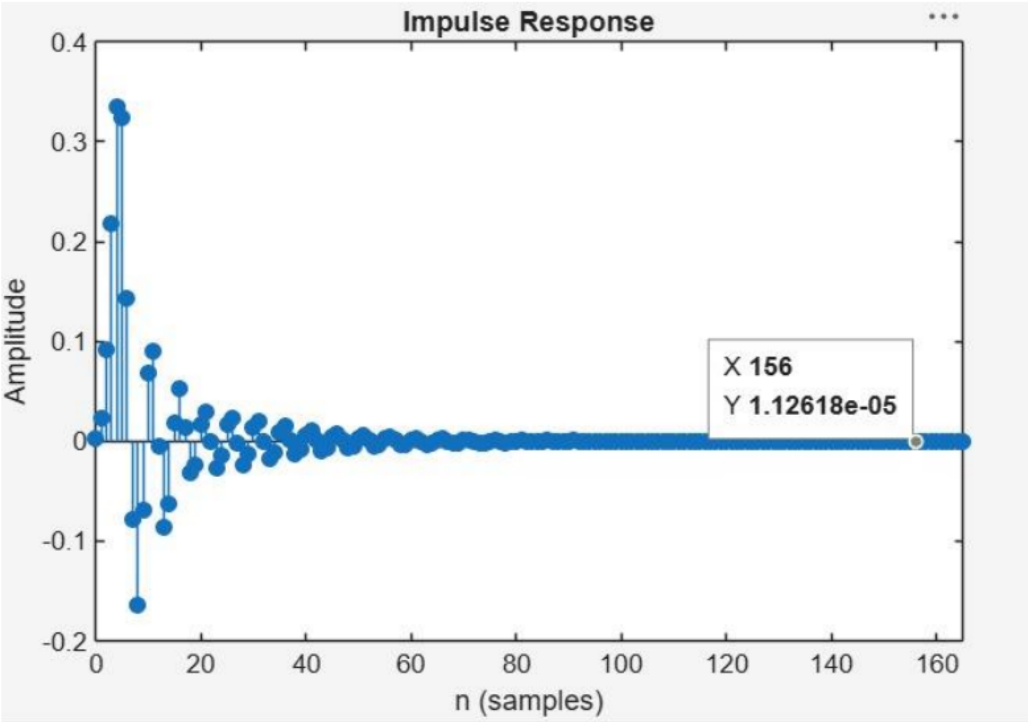
```
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;
//  NO:2
Chebyshev IIR Filter
// TE: 2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% -------- GRAPHS --------
figure; freqz(b,a);
figure; impz(b,a);
figure; grpdelay(b,a);
figure; zplane(b,a);
```

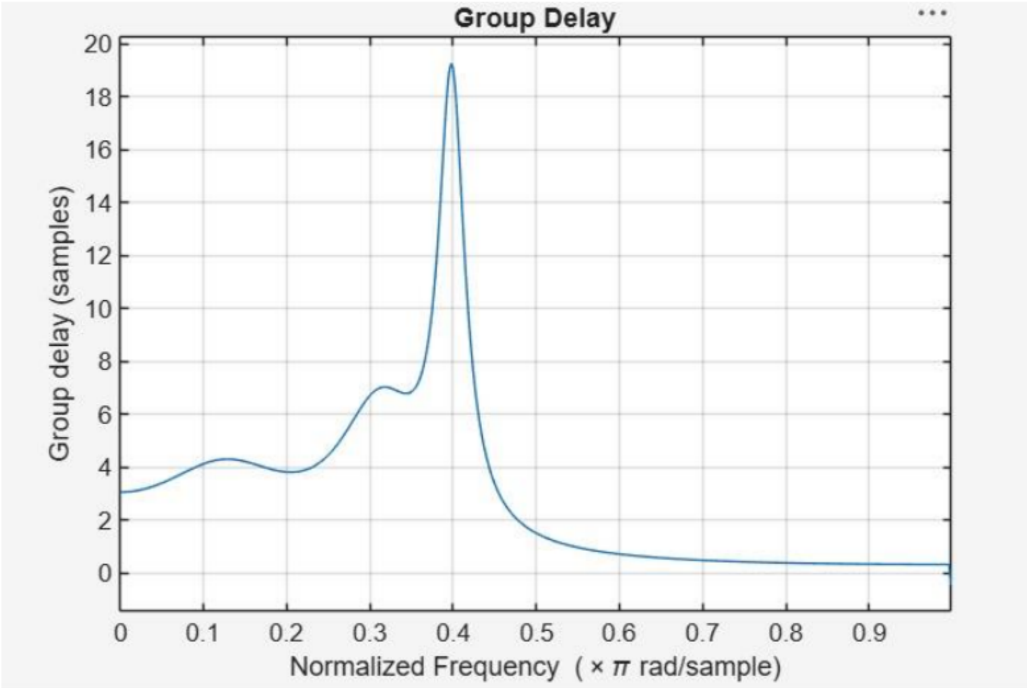
Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.



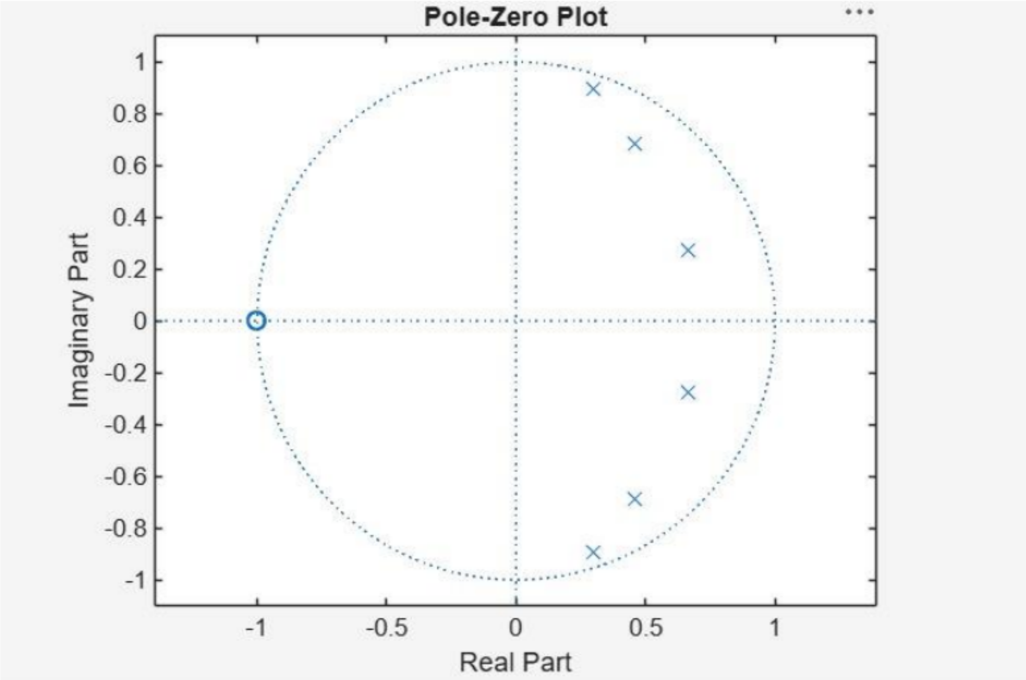
Objective 2:


Impulse Response
Amplitude
n (samples)
X 156
Y 1.12618e-05
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0
20
40
60
80
100
120
140
160



Group Delay
Group delay (samples)
20
18
16
14
12
10
8
6
4
2
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Normalized Frequency (× π rad/sample)

Objective 3:



Objective 4:

Filter Order = 6

Cutoff Frequency = 0.40 * pi rad/sample

System is STABLE

Result:

1. The Chebyshev Type-I IIR low-pass filter was successfully designed, and its magnitude response, phase response, impulse response, and group delay were analyzed.
2. The impulse response and group delay characteristics were obtained and found to satisfy the given specifications.
3. The pole-zero plot confirmed that all poles lie inside the unit circle, verifying the stability and proper performance of the filter.
4. The minimum filter order and cutoff frequency were successfully determined and found to satisfy the given design specifications.

Practical Question – 1

Design a digital **Chebyshev Type-I IIR Low-Pass Filter** using MATLAB to meet the following specifications: passband ripple of **1 dB**, minimum stopband attenuation of **40 dB**, passband edge frequency of **1000 Hz**, stopband edge frequency of **2000 Hz**, and sampling frequency of **8000 Hz**.

Task:

1. Determine the minimum filter order and cutoff frequency required to satisfy the given specifications.
2. Obtain and analyze the magnitude response, phase response.
3. Obtain and analyze impulse response, and group delay characteristics of the
4. designed filter. (iv) generate the pole-zero plot and verify the stability and performance characteristics of the filter.

Aim (10)	**Program (20)**	**Procedure/Calculations (20)**	**Results (20)**	**Record (10)**	**Viva (20)**	**Total (100)**

Practical Question – 2

Design a digital Chebyshev Type-I IIR Low-Pass Filter using MATLAB to meet the following specifications: passband ripple of 1 dB, minimum stopband attenuation of 50 dB, passband edge frequency of 2500 Hz, stopband edge frequency of 1500 Hz, and sampling frequency of 8000 Hz.

(i) Determine the minimum filter order and cutoff frequency required to satisfy the given specifications.
(ii) Obtain and analyze the magnitude response and phase response of the designed filter.
(iii) Obtain and analyze the impulse response and group delay characteristics of the designed filter.
(iv) Generate the pole-zero plot and verify the stability and performance characteristics of the filter.

Aim (10)	**Program (20)**	**Procedure/Calculations (20)**	**Results (20)**	**Record (10)**	**Viva (20)**	**Total (100)**

Result

The Chebyshev Type-I low-pass filter was successfully designed. The minimum order was obtained, the required responses were analyzed, and all poles were confirmed to lie inside the unit circle.

Student Details

Name: Dhinakaran. S
Reg. No.: 192412353
Course: ECA09 - Digital Signal Processing